

Bloch — Migration to Proof-of-Stake (SHA-3 + Lattice)

Project design document — "Genesis-4 / Bell"

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Fixed inputs (founder constraints). Two parts of the system are **not** up for redesign and are treated throughout as given:

1. **The PQ signature arrangement stays as it is** — the hybrid ML-DSA-65 | Falcon-1024 suite, unchanged, in every consensus role.
2. **The ZK ledger (Coherence) stays** — C1-frozen formats, SHAKE-256 primitives, raw FRI-STARK proofs, no elliptic-curve ZK.

These are constraints, not conveniences: the ≈ 4.6 KB hybrid signature is what sets slot time, committee size and block size (§5.1, §6.5), and keeping the shielded pool forces one new consensus rule on the staking design (§6.6.3). Both consequences are worked through rather than assumed away.

0. Read this first — the founder-level objection

This document designs, in full, a migration of the Bloch base chain from Proof-of-Work to Proof-of-Stake with SHA-3 hashing and lattice signatures. It is a complete, executable plan. It is also a plan whose **single largest risk is not technical**, and the PMO's duty is to state that before the architecture:

1. **Stake concentration.** As measured at height 18,809, one address (the founder) holds ≈ 3.427 B of 3.634 B existing BLCH — about **94% of circulating supply**. Proof-of-Stake makes coins the consensus weight. On day one of a naive PoS migration, the founder *is* the consensus. This is the exact inverse of the ownerless thesis restored in ADR-033 and of the "coins don't vote" governance model. Section 4.1 specifies consensus-level mitigations (ineligibility covenants, stake caps, Nakamoto-coefficient gates); none of them fully solve it, and §11 makes distribution a hard Go/No-Go gate rather than a best-effort.
2. **Securities framing.** Staking rewards paid to bonded holders materially strengthen an investment-contract reading of BLCH. The current public posture — a civic node movement, value beside the point, not a security — survives PoW far more comfortably than it survives PoS. Legal review is a **Phase 0 blocking deliverable**, not a Phase 5 checkbox.
3. **Discarded security budget.** The chain is today merged-mined with Bitcoin via AuxPoW (live since height 8,500). That borrowed SHA-256d hashrate is the cheapest real security Bloch will ever have. PoS discards it, and discards the "we court Bitcoin miners, not traders" go-to-market with it.

None of the three is a reason to refuse the work. They are reasons to run the plan with the gates in §11 armed, and to be honest in every public artifact about what is being traded away. The rest of this document assumes the founder has read §0 and still wants the migration designed and staffed.

1. Scope

1.1 In scope

- Replacement of SHA-256d with SHA-3 (SHA3-256 / SHAKE-256) as the chain's hash function across block identity, Merkle commitments, state commitments and randomness.
- Replacement of Proof-of-Work leader election with stake-weighted leader election plus a BFT finality rule.
- Lattice (post-quantum) signatures in a **consensus** role for the first time, using **the existing signature arrangement unchanged**: the hybrid ML-DSA-65 | Falcon-1024 suite already in `crates/bloch-crypto/src/crypto/mod.rs`.
- The staking lifecycle: deposit, activation queue, attestation duty, rewards, slashing, exit, withdrawal.
- The migration mechanics themselves: hybrid period, flag-day, rollback.
- Ecosystem migration: pool, stratum, explorer, wallet, L2, SDKs, RPC.

1.2 Out of scope

- Changes to the emission curve. Emission V3 (ADR-035, tail floor 60 BLCH at epoch 6) is treated as fixed input; PoS changes *who* receives issuance, not *how much* is issued.
- Changes to the transaction format, address format, or the eUTXO VM, beyond the two new transaction types in §7.
- **Changes to the PQ signature arrangement.** The hybrid suite `SUITE_MLDSA65_FALCON1024` (0x0001) is a fixed input. No new signature algorithm, no role split, no per-message scheme selection. PoS consumes the signature layer exactly as it exists. §6.2 and §6.5 are written around that constraint rather than around a convenient one.
- **Changes to Coherence (the ZK ledger).** The C1-frozen shielded-pool formats — SHAKE-256 note commitments, hash-derived nullifiers, the SHAKE-256 incremental accumulator, `check_spend`, and raw FRI-STARK verification with no Groth16/EC wrapper — are preserved byte-for-byte. §6.6 specifies what the migration must not break, and the one new consensus rule the shielded pool forces on the staking design.
- The premine vesting schedule (10,368,000-block cliff + 480 monthly tranches). Premine ineligibility is retired with the taint machinery — see §4.
- Any L2/EVM change beyond re-pointing its anchor to the new finality signal.

1.3 What this document does not claim

- It does not claim lattice hardness secures consensus. The lattice primitives here are **signatures** (ML-DSA, Falcon) — their security is FIPS 204/206 signature security, not a hardness assumption about block production. This is the same discipline the Module-SIS PoW crate already applies to itself.
- It does not claim the leader election is private. §6.3 chooses a *public* sortition on purpose and pays a DoS cost for it, because no standardised post-quantum VRF exists (§6.4).

- It does not claim PoS is more decentralised than PoW for Bloch specifically. Given §0.1, the honest default assumption is that it is less so, until the distribution gate in §11 is met.

2. Baseline — what Bloch is today

Grounded in the reference node (`~/dev/BlochSISPoW-project` , branch `g3-integration`):

Property	Today
Consensus	Proof-of-Work, GhostDAG BlockDAG, $k = 8$
PoW hash	SHA-256d over an 80-byte Bitcoin-compatible <code>MiningHeader</code> projection (<code>crates/bloch-crypto/src/core/mod.rs:~700</code>)
Merged mining	AuxPoW active since h8,500; Bitcoin work secures Bloch
Block target	30 s (<code>TARGET_BLOCK_TIME_SECS</code> , <code>tokenomics_v2.rs:78</code>)
Block header	<code>version</code> , <code>parents: Vec<[u8;32]></code> , <code>merkle_root</code> , <code>timestamp: u64</code> , <code>bits: u32</code> , <code>nonce: u64</code> (<code>core/mod.rs:652</code>)
Difficulty	ASERT-style, <code>bits</code> per block
Signatures	Hybrid ML-DSA-65 Falcon-1024 , suite-tagged envelope (<code>SUITE_MLDSA65_FALCON1024 = 0x0001</code>); pk 1,952 1,793 B, sig 3,309 ~1,280 B, both must verify (<code>crypto/mod.rs:16-42</code>)
Signature escape hatch	<code>SUITE_MLDSA65_ONLY = 0x0002</code> already defined — Falcon is removable without a format break
ZK ledger	Coherence shielded pool, C1-frozen: SHAKE-256 commitments/nullifiers, SHAKE-256 incremental accumulator (depth 32), SP1 raw FRI-STARK proofs, no elliptic-curve ZK (<code>crates/coherence-core</code> , <code>src/coherence</code> , wired in <code>main.rs:844</code>)
Hashing	SHA-2 for consensus, SHA-3/SHAKE-256 already a dependency (<code>sha3 0.10</code>) and already the native hash of Coherence, audit chains and L2 commitments
Finality	PoW depth only. FFG-BFT explicitly dropped (<code>docs/specs/POSTERN-FFG-BFT-FINALITY.md</code>)
Nominal supply	21 B BLCH, not hard-capped (perpetual tail)
Distribution	≈ 94% of existing supply at one address (§0.1)

Two known consensus fragilities inform the design and are called out where they bite:

- **Block-identity keying.** A prior defect keyed the DAG by `pow_hash` while storage keyed bodies by `block_hash` . Any change to the hash function walks straight through this area (§5.4).
- **Order-dependent difficulty validation.** `expected_bits` is derived from mutable local state rather than from ancestry, so nodes running identical binaries can diverge at retarget heights. PoS **deletes this bug class outright** (there is no difficulty), which is a genuine, non-marketing argument in the migration's favour — but §5.5 requires that stake state not reintroduce the same pattern.

3. Target architecture — summary

Name: Genesis-4, codename **Bell**. Merge event = "the Bell transition".

Layer	Choice	Rationale
Hash	SHA3-256 (fixed) + SHAKE-256 (XOF), domain-separated	Keccak sponge, no length-extension, PQ-adequate (Grover: quadratic only), already in-tree
Block identity	SHA3-256(DS_BLOCK canonical_header)	Replaces SHA-256d; §5.4
Leader election	Stake-weighted public sortition over a hash-based beacon	No standardised PQ VRF exists; §6
Randomness	Hash-based commit-reveal (RANDAO-style) over SHAKE-256 preimages, mixed per slot	Uniqueness comes from preimage binding, not signature uniqueness
All signatures	Unchanged: hybrid ML-DSA-65 Falcon-1024 suite, for identity, proposal and attestation alike	Founder constraint; no new primitive enters the tree
Finality	Bloch-BFT: two-round, $\geq 2/3$ stake, checkpoints every epoch	Deterministic finality; replaces PoW depth
Chain shape	Linear chain, 30 s slots , 32-slot epochs, committee 128 voting at the epoch boundary + 8 per slot	Cadence, not committee size, is what the 4.6 KB signature actually constrains; §6.5
Vote compression	Optional optimisation — epoch-boundary FRI-STARK proof over the quorum	Measurement removed the dependency; §6.5.1
ZK ledger	Coherence C1 preserved; its roots enter <code>state_root</code> ; shielding is closed to tainted coins	§6.6

4. Distribution and the anti-capture design — retired and replaced

This section described taint propagation, and taint no longer exists. It was written for a migration *in place*, where the founder's ~94% holding sat on the chain being converted and had to be marked ineligible coin by coin. Two later decisions removed its subject: Genesis-3 halts and Genesis-4 launches from a snapshot (BLOCH-TOKENOMICS-V4.md §3.2), and the carryover crosses as **one undifferentiated set with no exclusion list** (§1 of the same document). There is no class of coin left to mark.

Three mechanisms went with it: the taint set, the 300 M holder cap, and — the one worth naming — the **exclusion list as an unaudited power**. Whoever wrote that list would have decided who counts as founder, with nothing in the protocol checking it. That risk did not get mitigated; it stopped existing.

4.1 What answers concentration instead

Not coin marking. Three things, in descending order of how much they actually do:

1. **The allocation itself.** The founder's new grant is 17% under a 10-year cliff and 40-year linear vest — the strictest schedule on the chain, far beyond any market benchmark.

2. **Vesting on the Foundation buckets.** VC and team hold nothing liquid at genesis; marketing releases a quarter; liquidity is liquid by function (`BLOCH-TOKENOMICS-V4.md` §7B).
3. **Gates measured on stake that is not insider stake.** G1–G4 exclude stake whose beneficial owner is the Foundation, the founder, or Postern Labs — including Foundation-delegated stake and the genesis validator set (`BLOCH-ENTITY-STRUCTURE.md` §5.1, tokenomics §3.3).

4.2 What that does not fix, stated plainly

The carried-over balance is **liquid at genesis**, and the largest single address holds 3,546,175,400 BLCH — 70.4% of the circulating supply at slot 0. Gate G2 requires the largest holder under 25%, which this schedule does not reach until roughly **year five**.

An earlier draft cliffed the founder's entire position and bought a genesis where the founder held no spendable stake at all. Carrying the balance across liquid gave that up. The tokenomics document states the year-by-year figures (§4A); this document's job is only to record that the gates in §11 are what enforce it, and that they are not met at launch.

One distinction stays available and is still undecided: **liquid is not the same as stakeable**. A carried-over balance can be spendable while remaining ineligible to stake. Nothing about carrying it across liquid decides that it votes, and that decision is what determines whether the gates are reachable before year five.

5. Chain structure

5.1 Slots and epochs

Constant	Value	Note
<code>SLOT_DURATION_SECS</code>	30	Deliberately identical to today's PoW block target — see below
<code>SLOTS_PER_EPOCH</code>	32	16 min/epoch
<code>EPOCHS_PER_CHECKPOINT</code>	1	Justification per epoch; finality $\approx$ 32 min
<code>MIN_DEPOSIT_BLCH</code>	100,000	Sized so a validator set of $\sim$ 1,000 is reachable from realistic float
<code>MAX_VALIDATOR_STAKE</code>	1% of active stake	1% cap, resolved by fixed point — <code>delegation.rs</code>
<code>COMMITTEE_SIZE</code>	removed	Replaced by partitioning — see §6.5.3
<code>SLOT_SUBCOMMITTEE_SIZE</code>	removed	Same
Committee size	derived: $\text{ceil}(N / 32)$	The active set is partitioned, not sampled
<code>ATTESTATION_CADENCE</code>	once per validator per epoch	Each validator serves in exactly one slot committee
<code>ACTIVATION_DELAY_EPOCHS</code>	8	$\sim$ 2.1 h
<code>EXIT_DELAY_EPOCHS</code>	32	$\sim$ 8.5 h
<code>WITHDRAWAL_DELAY_EPOCHS</code>	2,048	$\sim$ 22.8 days; weak-subjectivity margin

Constant	Value	Note
SLASH_PROPOSER_EQUIV	5% of stake + ejection	Two blocks, same slot
SLASH_SURROUND_VOTE	5% of stake + ejection	Casper-style surround/double vote
INACTIVITY_LEAK_RATE	quadratic after 4 epochs of non-finality	Recovers liveness from a stalled set

All values are proposals for Phase 1 review; every one of them is a consensus constant requiring a KAT and a devnet sweep before it is frozen.

Why 30 s slots, a committee of 128, and epoch-boundary voting. Two earlier drafts got this wrong in opposite directions. The first used 12 s slots and a committee of 128, sized around a compact attestation signature the fixed arrangement does not provide. The second corrected the signature size but kept per-slot voting, and paid for it by halving the committee to 64.

Measurement settled it (`spikes/prover-cost/RESULTS.md`). The dominant variable is not slot time or committee size — it is **how often the committee votes**. Moving the full committee's vote to the epoch boundary cuts both proving cost and signature storage by 32×, which buys back the committee size *and* removes the need for aggregation. A small per-slot subcommittee (§6.5.2) preserves the fork-choice weight that epoch-only voting would otherwise destroy.

The remaining price is **finality latency** — ≈ 32 min, against ≈ 13 min for a 12 s-slot design. That is the honest cost of a 4.6 KB signature, and it is not recoverable without changing the signature arrangement.

5.2 Retiring GhostDAG

GhostDAG's blue-score ordering exists to make concurrent PoW blocks useful. With one designated proposer per slot, concurrency is an anomaly, not the norm. Genesis-4 is a **linear chain**:

- `BlockHeader.parents` narrows from `Vec<[u8;32]>` to a single `parent: [u8;32]` . The field is kept as a vector in the wire format with a consensus rule `parents.len() == 1` so that serialisation code, explorers and SDKs need a smaller diff.
- Fork choice becomes **LMD-GHOST over attestation weight**, with the latest justified checkpoint as the root — not blue score.
- `reachability.rs` and the DAG anticone machinery are retained but demoted to historical-chain queries only (pre-transition blocks remain a DAG forever).

Migration hazard: the historical DAG and the post-transition chain coexist in one database. The transition block is the unique block whose parent is a DAG selected-tip and whose children are linear. Every index, every RPC, and the explorer must handle that seam. Assistant A5 owns a seam test matrix.

5.3 Block header (Genesis-4)

```
BlockHeaderV4 {
    version:          u32           // 4 - 0xB10C_0005
    parent:           [u8; 32]      // 32 - SHA3-256 block id
    state_root:       [u8; 32]      // 32 - SHA3-256 of the eUTXO + stake state
    body_root:        [u8; 32]      // 32 - SHA3-256 Merkle root of transactions
    slot:             u64           // 8
    proposer_index:   u32           // 4 - index into the active validator set
    randao_reveal:    [u8; 32]      // 32 - SHAKE-256 preimage for this slot
    randao_mix:       [u8; 32]      // 32 - accumulated beacon after mixing
    justified_root:    [u8; 32]      // 32 - latest justified checkpoint
    finalized_root:    [u8; 32]      // 32 - latest finalized checkpoint
    attestation_root: [u8; 32]      // 32 - root over the quorum in the body
    coherence_root:   [u8; 32]      // 32 - Coherence accumulator root, §6.6
}

BlockEnvelope {
    header:      BlockHeaderV4
    proposer_sig: HybridSig      // ≈4,589 - ML-DSA-65 || Falcon-1024, suite-tagged
    body:        Body            // transactions + attestation quorum
}
```

Removed relative to V3: `bits`, `nonce`, `timestamp` (derived from `slot`), `parents: Vec<_>`. Removing `bits` is what deletes the order-dependent difficulty bug class described in §2.

The proposer signature sits in the **envelope, not the header**. With a 4.6 KB hybrid signature, an in-header signature would make the header itself larger than a typical Bitcoin block's worth of headers, and every light-client and header-sync path would carry it. Keeping the signed object small (`header` is 248 B) means header chains stay cheap and only full validation pays for the signature.

5.4 Block identity — the one change that touches everything

```
block_id = SHA3-256( DS_BLOCK || canonical_serialize(header_without_proposer_sig) )
```

`block_id` is the **only** identifier. There is no second hash, no `pow_hash`, and no mining projection. This is deliberate: the historical `pow_hash` / `block_hash` split caused the DAG-keying defect that stalled tip selection. Genesis-4 forbids the pattern at the type level — `BlockId` is a newtype with no second constructor and no `From<PowHash>`.

Assistant A2 owns a property test asserting that no code path in the tree derives a block identifier from anything other than `BlockId::of(&header)`.

5.5 State commitment and determinism

`state_root` commits to a SHA3-256 sparse Merkle tree over:

- the eUTXO set,
- the validator registry (pubkeys, stake, activation/exit epochs, slashed flag),
- the current and previous epoch attestation participation records,
- the randao mix history for the last 2 epochs,

- **the Coherence shielded-pool state:** the SHAKE-256 accumulator root and the nullifier-set root (§6.6). Today these live in an in-memory `ShieldedPool` (`main.rs:844`); under PoS they must be committed, because finality means nothing if the shielded state is not part of what gets finalized.

Hard rule, learned from the difficulty defect: every consensus-relevant value used to validate block *B* must be derivable from `B.parent`'s committed state, never from node-local mutable state. Validator set, proposer schedule, committee assignment and beacon mix are all read from `state_root` of the parent. A reviewer checklist item ("does this read local mutable state?") is a merge blocker for DEV-1, and A4 audits for it explicitly.

6. Cryptography

6.1 SHA-3 domain separation

One hash function, many uses; every use gets a tag. Tags are ASCII, fixed length 16, right-padded with `0x00`:

Tag	Use
<code>BLCH4:BLOCK\0...</code>	Block identity
<code>BLCH4:BODY\0...</code>	Transaction Merkle tree
<code>BLCH4:STATE\0...</code>	State SMT nodes
<code>BLCH4:ATTEST\0...</code>	Attestation signing root
<code>BLCH4:RANDAO\0...</code>	Beacon mixing
<code>BLCH4:SORTIT\0...</code>	Sortition draw
<code>BLCH4:DEPOSIT\0...</code>	Deposit message
<code>BLCH4:SLASH\0...</code>	Slashing evidence

Fixed-length digests use SHA3-256. Variable-length or multi-output derivation uses SHAKE-256. SHA-256d survives **only** in the historical verification path for pre-transition blocks, and is never used to produce anything new.

6.2 Lattice signatures — the arrangement is fixed

One suite, every role. Validator identity, deposits, exits, slashing evidence, block proposals and attestations all use the existing hybrid `SUITE_MLDSA65_FALCON1024`. There is no role split, no second scheme, and no per-message algorithm negotiation.

	ML-DSA-65	Falcon-1024	Hybrid total
Public key	1,952 B	1,793 B	3,745 B
Signature	3,309 B	~1,280 B	≈ 4,589 B
Verification	both must pass	both must pass	AND, not OR

What that buys and what it costs, stated plainly:

- **Buys:** no new primitive enters consensus. The code, the KATs, the wallet, the SDKs and the audit surface are the ones that already exist. A break in either lattice family alone does not break the chain — that is the whole point of the hybrid, and it is a genuinely strong position to hold into a consensus role.
- **Costs:** ≈ 4.6 KB per consensus message. Every sizing decision in §5.1 descends from that number, and §6.5 stops being an optimisation and becomes structural.

Falcon-1024 caveat, non-negotiable for DEV-2. Falcon signing uses floating-point Gaussian sampling with a documented history of side-channel and implementation pitfalls. PoS does not add Falcon to the tree — it is already there — but it **changes the threat model materially**: today Falcon signs occasionally, in a wallet, usually offline; under PoS it signs every slot on an internet-facing machine holding bonded stake. That is a different exposure for the same code. Requirements: constant-time / integer-emulated signing path only, no FP fallback in release builds, deterministic KATs, and remote-timing review by A4 against a validator that is signing on a schedule an attacker can predict (§6.4 makes the schedule public).

The escape hatch already exists and stays closed by default. `SUITE_MLDSA65_ONLY = 0x0002` is defined in-tree precisely to prove Falcon is removable without a format break. Under the founder constraint it is **not** adopted; it is documented here so that if A4's timing review returns a P0 on online Falcon signing, the response is a suite change already contemplated by the format — dropping to 3,309 B, 28% smaller — rather than an emergency redesign. Adopting it would be a founder decision reversing this constraint, recorded in the ADR.

6.3 Randomness beacon

RANDAO-style, hash-based, no VRF and no BLS:

1. At registration a validator commits `c_0 = SHAKE-256^k(seed)` for `k = 8,192` iterations (a hash chain).
2. In its assigned slot the proposer reveals `r_i`, the preimage one step down the chain. Nodes verify `SHAKE-256(r_i) == c_i` and set `c_{i+1} = r_i`.
3. `randao_mix_{n+1} = SHA3-256(DS_RANDAO || randao_mix_n || r_i)`.

Preimage binding gives the property ML-DSA cannot: **uniqueness**. There is exactly one valid reveal per slot, so a proposer cannot grind its contribution. It can *withhold* (skip its slot) — the standard RANDAO last-revealer bias, bounded at one bit of influence per withheld slot, and expensive because a skipped slot forfeits the proposer reward.

Chain exhaustion (after `k` slots) requires a re-commit transaction; A1 owns the exhaustion test.

6.4 Why not a VRF

The obvious design would be VRF-based private sortition (Algorand-style). It is not available post-quantum:

- **ML-DSA is not unique.** FIPS 204 signatures are randomised; even in hedged or deterministic mode, many valid signatures exist for one (key, message) pair. A hash of an ML-DSA signature is therefore **grindable** — a proposer can re-sign until the sortition output favours it. Any design that treats "hash of a PQ signature" as a VRF output is broken. This must be stated in the GIP so no future implementer reinvents it.
- **Falcon is not unique either**, and the hybrid is not unique twice over — so keeping the current arrangement does not open a VRF path that the role-split design lacked. The beacon in §6.3 is signature-

independent by construction, which is exactly why the founder constraint costs nothing here.

- **Lattice VRFs exist but are research-grade.** LB-VRF-style constructions (Esgin et al., 2021) achieve uniqueness only for a small bounded number of evaluations per key, requiring frequent key rotation. Interesting; not a foundation for mainnet in 2026.

Consequence, accepted deliberately: **sortition is public**. Anyone can compute the proposer schedule for the current epoch. Mitigations: proposer identity is an index, not a network address; validators are expected to run behind sentry nodes; the schedule is only revealed one epoch ahead. This is a real DoS surface and is documented as such rather than hidden.

A PQ-VRF track is kept open as research (§14) and can upgrade sortition later without touching the rest of the design.

6.5 Attestation footprint — the binding constraint

The hybrid signature is $\approx 4,589$ B, and the cadence decides everything. Measured figures, from `spikes/prover-cost/`:

Cadence	Signature bytes/year	Avg per block	In-circuit cycles/s
Committee 64, every slot	308.7 GB	286.8 KB	15.52 M
Committee 128, epoch only	19.3 GB	17.9 KB	0.97 M
Committee 128 epoch + 8 per slot (adopted, founder decision 2026-08-10)	57.9 GB	53.8 KB	2.91 M

Per-slot voting made the quorum the dominant term in block size and forced the committee down to 64. Epoch-boundary voting removes that pressure entirely: the chosen design carries a **larger** committee at **one fifth** the proving cost and **one fifth** the storage of the per-slot design.

6.5.1 Epoch aggregation — optional optimisation, not a dependency

An earlier draft of this section called STARK vote-aggregation **structural**, on the reasoning that without it the committee is pinned at 64 and the chain accretes ~ 310 GB/year of undiscardable signatures.

Measurement retired that claim. The 310 GB/year was a consequence of per-slot voting, not of the signature size. At epoch cadence the same committee — doubled to 128 — produces 19.3 GB/year, and the chosen design (§6.5.2) 57.9 GB/year. That is ordinary, prunable-at-leisure data. Aggregation would still be *nice* (it would let the committee grow without bound), but nothing in the design now **depends** on a research-frontier result. The spike did not find a way to pay the cost; it found a way not to incur it.

The design is a natural fit for infrastructure that already exists for the ZK ledger:

1. Per-slot attestations are carried raw in the block, as above. Nothing is proven in real time — no proving system is asked to keep up with a 30 s slot.
2. At each epoch boundary, a prover produces **one FRI-STARK proof** that the epoch's quorum signatures verified and that the checkpoint was justified by $\geq 2/3$ stake. This reuses `crates/coherence-prover` (SP1, Plonky3, raw FRI) and inherits its non-negotiable rule: **raw FRI verification, never SP1's Groth16 wrapper** — a curve SNARK would silently destroy the post-quantum property of the whole consensus (`COHERENCE-C1.md §3`).

3. Once an epoch is finalized and its proof is verified, **the raw signatures become prunable**. Archival nodes may keep them; a fully validating node syncing from a finalized checkpoint verifies the proof instead.

Measured cost of the in-circuit statement (`spikes/prover-cost/RESULTS.md` , 2026-08-10), on real PQClean signatures verified by pure-Rust verifiers compiled to `riscv32im`:

	Instructions	Keccak permutations	Keccak share
ML-DSA-65 verify	5,909,451	208	57.7%
Falcon-1024 verify	1,312,901	31	38.4%
Hybrid, marginal	7,274,849	239	54.2%

Two findings worth carrying forward. **Falcon-1024 verifies 4.5× cheaper than ML-DSA-65** — the bottleneck is the NIST-standardised half, not the exotic one, which inverts the intuition about which half to drop if cost ever forces the question. And **the cost is exactly linear in N** (0.41% spread across four distinct signature pairs, zero fixed overhead), so batching buys nothing in the guest: any saving must come from the proof system itself.

6.5.3 Partition, do not sample — the F1 correction

The adversarial review found that the sampled design has **no coherent quorum denominator**, and that both readings fail:

Denominator	Failure
Network stake	A 128-validator sample cannot hold 2⁄3 of network stake past ~192 validators — and gate G4 <i>requires</i> ≥ 200. Finality structurally unreachable ; the inactivity leak fires forever
Committee stake	A 128-sample has enough variance that a ~30%-stake adversary exceeds 1⁄3 of the committee in roughly one epoch in five, stalling finality below the nominal threshold

The fix is to partition the active set rather than sample it. Shuffle deterministically, cut into 32 committees, one per slot. Every validator lands in exactly one committee and votes exactly once per epoch, so the union of an epoch's committees *is* the active set. The denominator is then total active stake, unambiguous and reachable by construction — the property Ethereum has, which the sampled design gave up without noticing what it was buying.

It also removes finding **F2**. Under independent per-slot draws a validator was routinely drawn in several slots of one epoch and emitted several attestations sharing a `target_epoch` , which `is_double_vote` correctly flags as slashable. Honest validators slashed themselves, and the reward was harvestable. Under a partition, two attestations sharing a target epoch really are equivocation.

Cost, stated honestly. One hybrid signature per active validator per epoch, against 384 under the sampled design:

Validators	Sampled	Partitioned	Per slot	KB/slot
200	384	200	7	31
384	384	384	12	54
1,000	384	1,000	32	143

Validators	Sampled	Partitioned	Per slot	KB/slot
4,096	384	4,096	128	574

Partitioning is **cheaper below 384 validators** — where gate G4 puts the launch — and more expensive above. The scaling ceiling is around 4,096 validators, past which sub-sampling would have to return and F1 with it. Aggregation would lift the ceiling; the measured in-circuit cost (§6.5.1) says that is research, not engineering. Recording the ceiling now is better than discovering it at 5,000 validators.

6.5.2 Per-slot subcommittee — superseded by §6.5.3

Epoch-only voting removes the per-slot attestation weight that LMD-GHOST uses for fork choice. Without it, intra-epoch reorgs become cheap: ordering inside an epoch would rest on slot number and the proposer signature alone.

Ethereum does not actually do this. There, every validator votes once per epoch, but the set is sliced into 32 committees — one per slot — so every slot still carries attestation weight. Literal epoch-only voting is a different and weaker thing.

The design therefore keeps a **SLOT_SUBCOMMITTEE_SIZE = 8** sample attesting each slot purely for fork-choice weight, with the full 128 voting at the epoch boundary for justification and finality. Cost: 2.91 M cycles/s and 57.9 GB/year — against 15.52 M and 308.7 GB for the per-slot design it replaces.

6.6 Coherence (the ZK ledger) — preserved, and what PoS must not break

Coherence is kept exactly as frozen in `COHERENCE-C1.md`. Notably, it is the part of the system that needs **no cryptographic migration at all**: its commitments, nullifiers and accumulator are already SHAKE-256, and its proof system is already hash-based FRI with elliptic-curve ZK explicitly excluded. The SHA-3 migration in §6.1 brings the rest of the chain *to where Coherence already is*, rather than moving Coherence anywhere.

Four requirements follow.

6.6.1 Continuity across the transition

The commitment accumulator is **incremental** and the nullifier set is **monotone**. Neither may be reset, re-rooted, or rebuilt at `TRANSITION_HEIGHT`. A note shielded at height $h < \text{transition}$ must be spendable at $h + 1$ with a witness computed under the old tree, and every nullifier ever published must remain permanently unspendable. This is a correctness requirement with a privacy failure mode: a reset accumulator would force every holder to re-shield, linking old notes to new ones and retroactively de-anonymising the pool.

A3 owns a shadow-fork test that shields before the transition and spends after it, asserting witness validity and nullifier persistence across the seam.

6.6.2 Shielded state must be finalized state

The shielded pool is currently in-memory (`ShieldedPool::new()`, `main.rs:844`). Under PoS its accumulator root and nullifier-set root are committed in `state_root` and mirrored in `BlockHeaderV4.coherence_root` (§5.3, §5.5). Finality that does not cover the shielded pool would leave the private ledger reorganisable after the transparent one is settled — the worst possible asymmetry, and precisely the class of divergence this chain has already been bitten by when consensus state lived outside the committed state.

6.6.3 Deposits must spend transparent outputs

Stated as consensus rules, enforced by every node:

```
INVALID deposit_tx if any input is a shielded output
```

The companion rule — that shielding a tainted coin is invalid — went with the taint set (§4). What remains is the direction that was always load-bearing on its own: **stake must be attributable**, so a validator's bond always traces to transparent coins.

The first rule is enforceable because a shield transaction spends **transparent** inputs, whose ancestry is public — the privacy boundary is not violated by checking it. The second rule closes the reverse direction: stake must be attributable, so a validator's bond always traces to transparent coins.

The two-class-coin cost this rule used to carry is gone with the taint set: no coin is marked, so none is second-class. The remaining constraint is narrow and uncontroversial — a shielded note has no `OutPoint` to bond, so it could not back a deposit in any case.

A9's audit found this is not a retrofit at all: the shield bridge **does not exist yet**. `ShieldedTx` has no transparent fields and `check_spend`'s balance equation means value can neither enter nor leave the pool. So the rule is a day-one design constraint on a bridge that has still to be built, at no privacy cost and no migration cost.

6.6.4 Shared prover infrastructure

`crates/coherence-prover` (SP1 guest + host + HTTP service, GPU-deployed) is the same infrastructure §6.5.1 needs for epoch aggregation. One prover service, two statements. This is a real economy of scope — and a real coupling: an outage or a regression in that service now affects consensus pruning as well as shielded spends. A6 owns availability requirements; the consensus path must degrade to "keep raw signatures" rather than stall when the prover is unavailable.

7. Staking lifecycle

Two new transaction types in the eUTXO model.

7.1 DEPOSIT

```
DepositTx {
  amount:          u64          // ≥ MIN_DEPOSIT_BLCH, ≤ MAX_VALIDATOR_STAKE
  validator_pubkey: [u8; 3745]  // suite-tagged hybrid: ML-DSA-65 || Falcon-1024
  randao_commitment: [u8; 32]   // c_0, §6.3
  withdrawal_addr:  Address     // where the stake returns
  proof_of_possession: HybridSig // ≈4,589 B over SHA3-256(DS_DEPOSIT || fields)
}
```

Validity: inputs must be **transparent** (§6.6.3); amount within bounds; PoP valid under **both** halves of the suite; suite tag == `SUITE_MLDSA65_FALCON1024`. One key pair serves identity, proposal and attestation — there is no separate attestation key, because there is no separate attestation algorithm. The deposit output is locked to a consensus-owned script; it is not spendable until withdrawal.

Deposits are accepted **on the PoW chain during the hybrid phase** — this is how the validator set exists before the transition.

7.2 EXIT and withdrawal

Voluntary exit is a hybrid-signed message; the validator stops being assigned after `EXIT_DELAY_EPOCHS`, and the stake becomes spendable to `withdrawal_addr` after `WITHDRAWAL_DELAY_EPOCHS`. The long withdrawal delay is the weak-subjectivity margin: it must exceed the window in which an exited validator could sign a conflicting history for free.

7.3 Slashing

Evidence transactions carry two conflicting signed messages. Penalty: `SLASH_*` percentage plus immediate ejection; a whistleblower reward of 1/32 of the penalty goes to the including proposer. Correlated-slashing amplification (penalty scaled by how much stake was slashed in the same window) is **included** — without it, a single entity controlling many validators is punished no more than an unlucky solo operator.

7.4 Rewards

Issuance per Emission V3 is unchanged in *quantity*. The recipient split becomes: 7/8 to attesters pro-rata to participation, 1/8 to the proposer, with the inclusion-delay weighting standard to Casper-style designs. Transaction fees follow the existing endowment split (`ENDOW_FEE_SHARE_BPS = 1000`).

`MINER_SHARE_BPS / VALIDATOR_SHARE_BPS` in `tokenomics_v2.rs:119–121`, today 100/0, invert at the transition. Note that this file currently asserts `VALIDATOR_SHARE_BPS == 0` with the comment "removed (no BFT)" — the constants and their comments are consensus-adjacent documentation and must be corrected in the same commit, not later.

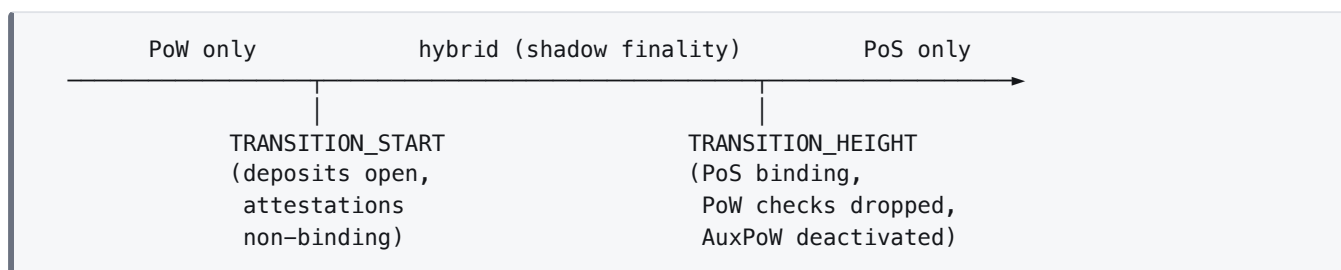
8. The transition mechanism

SUPERSEDED 2026-08-10. This section designed Genesis-4 as a flag-day fork on the running chain, with a hybrid period, and explicitly rejected a fresh genesis. The founder decided otherwise: the chain **halts** at height 80,000, a signed balance snapshot is taken, and Genesis-4 launches from it about six months later after code review (`BLOCH-TOKENOMICS-V4.md §3.2`).

What that changes: there is no hybrid period, no `TRANSITION_START`, no DAG→linear seam to cross, and no upgrade partition — the old chain is not continued, it is ended. §4.1's taint machinery goes with it: it was built because the live distribution could not be fixed in place, and a fresh genesis replaces coin-marking with the allocation and its vesting schedules.

The text below is retained because the gates (§11), the rollback thinking (§13) and the testing strategy (§12) still apply to launching a new chain.

Genesis-4 was designed as a **flag-day fork on the existing chain**, not a new genesis. The carryover approach used for Genesis-3 was rejected here: a new genesis at that stage would reset the very distribution history that §4.1 depends on.



- **TRANSITION_START** (height, published ≥ 8 weeks ahead): deposit transactions become valid; validators begin attesting; attestations are **gossiped, stored, and scored but do not affect fork choice**. The chain is still pure PoW. This is a live-fire rehearsal on mainnet with zero consensus authority — the single most valuable de-risking step in the plan.
- **TRANSITION_HEIGHT**: activation, gated by §11. From this block, PoW validity checks are dropped, `AUXPOW_ACTIVATION_HEIGHT` is superseded, and fork choice switches to LMD-GHOST over attestations. The transition block itself is the last PoW block and the first finalized checkpoint.
- **Nodes that do not upgrade** stop following the chain at `TRANSITION_HEIGHT` — they will see the first V4 block as invalid. This is a clean partition, not a silent divergence, and is the correct behaviour.

Given the yamux/release-gap history in this project, **release engineering is a first-class risk**: the published binary must be the binary the fleet runs, and the fleet must be verified on it before the flag day, not during. Assistant A6 owns this and holds a veto (§9.4).

9. Organisation

PMO + 3 developers (Claude Fable 5) + 6 assistants.

9.1 PMO

Owens the plan, the risk register, the gates (§11), the flag-day calendar, GIP and ADR filing, and the external comms sequence (English, per standing policy). The PMO does not write consensus code. The PMO's core deliverable is that no phase exits without its exit criteria met and evidenced.

9.2 Developers

Role	Owens	Primary deliverables
DEV-1 — Consensus core	Slots/epochs, fork choice, finality, state transition, transition mechanism	LMD-GHOST + justification/finalization, <code>BlockHeaderV4</code> , DAG→linear seam, flag-day activation logic
DEV-2 — Cryptography, beacon & prover	SHA-3 domain separation, hybrid-suite integration, RANDAO, sortition, epoch aggregation	Hash migration across the tree, constant-time Falcon-1024 signing path, beacon + sortition + full KAT suite, the §6.5.1 proving-cost spike
DEV-3 — Staking, economics, node, P2P & ZK-ledger continuity	Deposit/exit/withdrawal, slashing, rewards, gossip, sync, RPC, wallet, Coherence state commitment	Two new tx types, the §6.6.3 transparent-input rule, reward accounting, attestation gossip topics, RPC surface, sync from a finalized checkpoint, shielded roots in <code>state_root</code>

Interfaces between the three are frozen at the end of Phase 1 as Rust traits with no implementations; every later phase codes against them.

9.3 Assistants

#	Assignment	Standing deliverable
A1	Conformance & test vectors	Cross-implementation KATs for every hash tag, signature, beacon step, sortition draw; the vector file is the spec's ground truth
A2	Fuzz & property testing	Fuzz targets for header/tx/attestation decode; property tests for the §5.4 identity invariant and §5.5 determinism rule
A3	Devnet & fleet operations	Multi-node devnets, shadow-fork replays of mainnet, chaos scenarios (§12), fleet health during hybrid
A4	Adversarial review & security	Threat model, per-phase adversarial pass, long-range/weak-subjectivity analysis, slashing-evidence review, §6.4 grinding review
A5	Documentation & ecosystem	GIP-0003, ADR-036, spec upkeep, explorer/wallet/SDK/L2 migration, the DAG-seam test matrix, public-facing English copy
A6	Release engineering	Reproducible builds, prebuilt fleet binaries, upgrade runbooks, rollback packaging, "published == running" verification

9.4 Rules of engagement

- **A4 and A6 hold vetoes.** A4 can block a phase exit on an unresolved security finding; A6 can block a flag day on a build/fleet mismatch. Only the founder overrides, and the override is recorded in the ADR.
- **Two-reviewer rule on consensus code.** Every consensus-affecting merge needs one other DEV plus A4.
- **No consensus constant lands without a KAT** (A1) and a devnet sweep (A3).
- Weekly: PMO risk-register review. Per phase: written exit-criteria evidence.

10. Phases

Phase 0 — Decision and framing (weeks 1–3) — BLOCKING

Deliverables: this document reviewed and accepted or rejected by the founder; legal review of the securities framing (§0.2); GIP-0003 filed; ADR-036 filed recording the reversal of the ownerless-PoW position; a public English statement of what is being traded away (merged mining, Bitcoin-miner GTM).

Exit: founder go/no-go recorded. **If legal review is negative, the project stops here.** Nothing downstream is worth doing under an unresolved framing risk.

Phase 1 — Specification freeze (weeks 4–9)

Full consensus spec: constants, header, state layout, fork choice, finality, sortition, staking lifecycle, slashing. Interface traits frozen. Threat model v1 (A4). KAT format defined (A1).

Exit: spec reviewed by all three DEVs plus A4 with no open P0 questions; every constant in §5.1 either confirmed or revised with rationale.

Phase 2 — Cryptographic foundation (weeks 6–13, overlaps Phase 1)

SHA-3 domain-separated hashing across the tree; constant-time Falcon-1024 signing path with KATs; RANDAO chain; sortition; hybrid deposit/PoP path; **the §6.5.1 epoch-aggregation proving-cost spike**. DEV-2 leads, A1 and A2 in lockstep.

Exit: full KAT suite green; Falcon signing path reviewed by A4 with no unresolved side-channel finding under a predictable signing schedule; fuzzers clean for 72 h; **a measured proving cost for one epoch's quorum, with the aggregation go/no-go decision recorded** (§6.5.1).

Phase 3 — Consensus implementation (weeks 10–19)

DEV-1: slots, epochs, LMD-GHOST, justification/finalization, `BlockHeaderV4`, state transition, DAG→linear seam. DEV-3: deposits, exits, slashing, rewards, gossip, RPC, sync-from-checkpoint. A3 runs devnets continuously from week 12.

Exit: a 30-node devnet finalizes for 7 consecutive days including induced partitions, 1/3 offline, and equivocation injection; all slashing paths exercised.

Phase 4 — Shadow fork and public testnet (weeks 18–25)

Shadow-fork mainnet state at a recent height and run the transition on it, repeatedly. Then a public testnet ("Bell testnet") with external operators — external participation is the point; a testnet run only by Postern proves nothing about the validator-set assumption.

Exit: three consecutive successful shadow-fork transitions, **each including a shield-before / spend-after Coherence continuity test** (§6.6.1); testnet finalizing for 14 days with ≥ 20 independent external validators; sustained block propagation measured at the 296 KB/block working point (§6.5) with no mesh or stream-limit regressions.

Phase 5 — Mainnet hybrid (weeks 24–?, duration set by §11)

`TRANSITION_START` on mainnet. Deposits open. Attestations non-binding. Distribution program runs (§4.3). Fleet on the release binary, verified (A6).

Exit: §11 gates all green. **This phase has no fixed end date by design** — it ends when distribution allows, or the project halts.

Phase 6 — Transition and post-merge (flag day + 8 weeks)

`TRANSITION_HEIGHT`. Then: withdrawals enabled after `WITHDRAWAL_DELAY_EPOCHS`; explorer/wallet/L2 anchor migration completed; AuxPoW and stratum decommissioned. Epoch aggregation (§6.5.1) is **not** on this critical path — the cadence change already delivered what it was for; it returns only if the committee needs to grow well beyond 128.

11. Go/No-Go gates for `TRANSITION_HEIGHT`

Every one must be green. Any red halts the flag day; there is no partial activation and no "we'll fix it after the merge".

#	Gate	Threshold
G1	Independent eligible stake	$\geq 15\%$ of circulating supply is deposited by parties that are not the Foundation, the founder, or Postern Labs
G2	Concentration	No single entity controls $> 25\%$ of active stake; top-3 $< 50\%$
G3	Nakamoto coefficient	≥ 7 for block production and for finality
G4	Validator count	≥ 200 active validators, ≥ 50 operated by parties unaffiliated with Postern Labs
G5	Client diversity or explicit acceptance	Single-client risk formally accepted in writing by the founder if no second implementation exists
G6	Hybrid stability	≥ 30 days of non-binding attestations with $\geq 95\%$ participation and zero unexplained divergence
G7	Security	A4 sign-off; no open P0/P1; external review of the online Falcon-1024 signing path
G8	Release integrity	A6 sign-off: published binary == fleet binary, reproducible, rollback package staged and tested
G9	Legal	Phase 0 legal position still valid under the final reward design
G10	Network capacity	54 KB/block average and the epoch-boundary burst (≈ 588 KB) sustained on the real fleet for ≥ 14 days: no gossip-mesh degradation, no yamux stream-limit failures, p99 propagation < 5 s
G11	ZK-ledger continuity	Shield-before / spend-after passes on three shadow forks; nullifier set and accumulator provably unbroken; shielded roots finalized (§6.6)

G1–G4 are the honest statement of §0.1: **if the coins do not distribute, the migration does not happen.** That is the correct outcome, not a failure of the engineering.

12. Testing, verification, chaos

- **KATs (A1):** every hash tag, every signature op, beacon steps, sortition draws, deposit/exit/slash message encodings, state-root computations. The vector file is normative for any future implementation.
- **Property tests (A2):** block identity has exactly one derivation (§5.4); no consensus value reads node-local mutable state (§5.5); state root is deterministic under transaction reordering within a block; serialisation round-trips.
- **Fuzzing (A2):** header, transaction, attestation, deposit, slashing evidence decoders; differential fuzzing of SHA-3 tag handling.
- **Chaos scenarios (A3):** 1/3 offline; 1/3 + 1 offline (inactivity leak recovery); network partition and heal; equivocating proposer; mass simultaneous exit; beacon-chain exhaustion; clock skew ± 30 s; a validator set that is 60% one operator.
- **Shadow forks (A3):** repeated transitions from real mainnet state — the only test that exercises the DAG→linear seam against real data, and the only place the Coherence continuity test (§6.6.1) is meaningful.
- **ZK-ledger tests (A1/A3):** shield before the transition, spend after; nullifier double-spend attempts across the seam; a shield transaction with a deposit spending a shielded output must be rejected (§6.6.3);

shielded roots present and finalized in `state_root`.

- **Adversarial passes (A4):** one per phase, plus a dedicated long-range / weak-subjectivity analysis and a sortition-grinding analysis (§6.4).

13. Rollback and abort

- **Before `TRANSITION_HEIGHT`:** abort is free. The hybrid phase changes nothing binding; deposits are refundable by a consensus rule that unlocks deposit outputs if the transition does not occur by `TRANSITION_DEADLINE_HEIGHT`. **This refund rule ships in the same release as the deposit rule** — it is not added later.
- **After `TRANSITION_HEIGHT`:** rollback is a coordinated hard fork back to PoW at a chosen height, discarding post-transition history. It is catastrophic, it requires the ASIC fleet and pool to be recoverable on short notice, and A6 stages the PoW binary + runbook for the entire post-merge window. Treat it as a fire extinguisher, not a plan.
- **Abort triggers:** any G-gate turning red pre-flag-day; a P0 cryptographic finding at any time; finality stalling > 4 epochs on the hybrid rehearsal; negative legal position.

14. Open questions

1. **PQ-VRF.** Can a bounded-evaluation lattice VRF (LB-VRF family) with epoch-scoped key rotation give private sortition at acceptable cost? Would upgrade §6.3–6.4 without touching the rest.
2. **STARK epoch aggregation (§6.5.1).** ~~Highest-value unknown~~ — **answered and demoted.** The in-circuit cost was measured (7.27 M instructions per hybrid verification, linear in N) and the cadence change removed the dependency. What remains open is only the upside case: if the committee ever needs to exceed ~128, what does one FRI proof over the epoch quorum cost on affordable hardware? Requires the SP1 toolchain to convert cycles into seconds.
3. **Falcon-1024 online.** Is a constant-time integer-only Falcon signer available and reviewable for a machine that signs on a publicly predictable schedule? The arrangement is fixed, so the answer here is a *security* question, not a design choice — and if it is P0, §6.2's suite escape hatch is the only lever, and pulling it requires reversing the constraint.
4. ~~Taint set permanence and the two-class coin.~~ **Dissolved.** The carryover crosses as one undifferentiated set, so no coin is marked and none is second-class. What replaced the question is narrower and still open: whether a carried-over balance that is *liquid* is also *stakeable* (§4.2). That one decides whether the gates are reachable before year five.
5. ~~Weak subjectivity — who signs the checkpoint in an ownerless system?~~ **Answered by ADR-036:** the ownerless premise is retracted and the Foundation publishes checkpoints. The question that replaces it is narrower and practical — publish under an *m-of-n* key held beyond the Foundation, with an explicit review date, so the arrangement does not become permanent by inertia (`BLOCH-ENTITY-STRUCTURE.md` §5.3). The mechanism — checkpoint format, cadence, boot consumption, and the m-of-n/review-date parameters — is specified in `BLOCH-WEAK-SUBJECTIVITY.md`.

6. ~~Do nothing — fix the difficulty defect and stay on PoW.~~ **Overtaken by events.** The founder decided on a Genesis-4 relaunch with new tokenomics on 2026-08-10, so the live chain is not the thing being changed; it is being replaced. This option is retained only as history — it was the right question to ask while the alternative was amending the running chain.
7. **Attestation cadence** — ~~per-slot vs epoch-boundary voting~~ **decided:** the hybrid, 8 per slot for fork-choice weight plus 128 at the epoch boundary for finality (§6.5.2). Confirmed 2026-08-10.

15. Immediate next actions

#	Action	Owner	Gate
1	Founder decision on §0	Founder	Phase 0
2	Legal review of PoS reward framing	PMO	Phase 0, blocking
3	File GIP-0003 and ADR-036	A5	Phase 0
4	Measure the untainted float precisely at current tip	DEV-3	Feeds G1
5	Falcon-1024 constant-time signer availability spike	DEV-2	Phase 1
6	Freeze consensus interface traits	DEV-1	Phase 1 exit
7	Draft the distribution program (§4.3)	PMO	Phase 5 prerequisite
8	Epoch-aggregation proving-cost spike (§6.5.1)	DEV-2	Phase 2 exit
9	Founder decision on the two-class coin (§6.6.3, §14.4)	Founder	Phase 1

16. Copyright

Released under MIT OR Apache-2.0, consistent with the reference node.

Appendix A — Constant summary

Constant	Proposed	Section
SIGNATURE_SUITE	0x0001 ML-DSA-65 Falcon-1024 (unchanged)	6.2
SLOT_DURATION_SECS	30	5.1
SLOTS_PER_EPOCH	32	5.1
MIN_DEPOSIT_BLCH	100,000	5.1
MAX_VALIDATOR_STAKE	1% of active stake	4.1
MAX_ACTIVATIONS_PER_EPOCH	4	4.1
COMMITTEE_SIZE	128 (voto na fronteira de época)	5.1, 6.5

Constant	Proposed	Section
SLOT_SUBCOMMITTEE_SIZE	8 (só peso de fork-choice)	6.5.2
ACTIVATION_DELAY_EPOCHS	8	5.1
EXIT_DELAY_EPOCHS	32	5.1
WITHDRAWAL_DELAY_EPOCHS	2,048	5.1, 7.2
SLASH_PROPOSER_EQUIV	5% + ejection	5.1, 7.3
SLASH_SURROUND_VOTE	5% + ejection	5.1, 7.3
RANDAO_CHAIN_LENGTH	8,192	6.3
BLOCK_VERSION	0xB10C_0005	5.3

Appendix B — Ecosystem impact checklist (A5)

Component	Impact
Coherence shielded pool	Formats unchanged ; roots move into committed state; shield closed to tainted inputs (§6.6)
coherence-prover (SP1/FRI service)	Second consumer: epoch aggregation (§6.5.1); availability now consensus-adjacent
Wallet shielded flows	Must surface "this coin cannot be shielded / staked" for tainted outputs
bloch-pool, stratum V1/V2, proxies	Decommissioned at flag day
AuxPoW / merged mining	Deactivated; BTC submit path retired
ASIC fleet (S19j Pro, 100 TH/s)	No longer usable for Bloch; disposition decision needed
Explorer (blochl1.com)	New header fields, finality display, DAG→linear seam
Wallet / PWA	Deposit/exit UX, staking views
L2 (bloch-l2-evm, chainId 8400)	Anchor to finalized checkpoints instead of PoW depth
SDKs (Python, Go), OpenAPI	Header schema, new RPCs
Node operators	Mandatory upgrade; clean partition if they do not
Snapshot / onboarding runbook	Sync-from-finalized-checkpoint replaces k=8 DAG snapshot